Lecture (5)

Properties of elements and the periodic table

The periodic table: Periodic table is a chart of the elements arranged into rows and columns according to their physical and chemical properties.

The periodic table of the chemical elements: is a tabular display of the chemical elements. The invention of the periodic table is generally credited to Russian chemist Dmitri Mendeleev who intended the table to illustrate recurring ("periodic") trends in the properties of the elements.

Mendeleev's and the German chemist Lothar Meyer. independently published their periodic tables in 1869 and 1870, respectively. They both constructed their tables in a similar manner by listing the elements in a row or column in order of atomic weight and starting a new row or column when the characteristics of the elements began to repeat. He left gaps in the table when it seemed that the corresponding element had not yet been discovered.

Modern periodic table:

i. It was proposed by Henry Moseley.

ii. Modern periodic table is based on atomic number.

iii. Moseley did an experiment in which he bombarded high speed electron on different metal Surfaces and obtained X-rays. He found out that $\sqrt{v}\ a\ Z$ where v = frequency of X-rays. From this experiment, Moseley concluded that the physical and chemical properties of the elements are periodic function of their atomic number.

Modern periodic Law – The physical and chemical properties of elements are a periodic function of the atomic number.

The periodic table had listed the elements in order of increasing atomic number.

A group or family: is a vertical column in the periodic table.

Groups are considered the most important method of classifying the elements have similar chemical properties.

There are 18 groups in the standard periodic table, including the d-block elements, but excluding the f-block elements. According to IUPAC 18 vertical columns are named as 1st to 18th group.

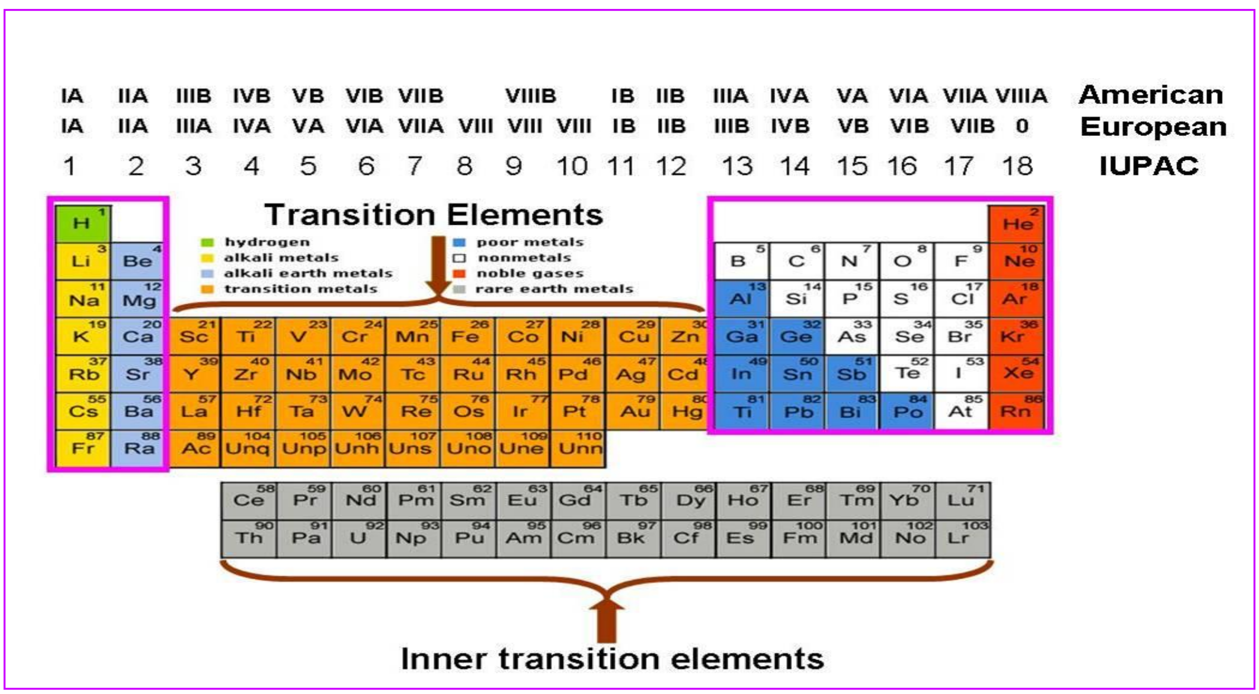
1A	IIA	IIIB	IVB	VB	VIB	VIIB		VIIIB		IB	IIB	IIIA	IVA	VA	VIA	VIIA	0
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

- alkali metals, members of Group 1A—lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), and francium (Fr)—are veryactive elements that readily form ions with a +1 charge when they react with nonmetals.
- alkaline earth metals The members of Group 2A—beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), and radium (Ra). They all form ions with a +2 charge when they react with nonmetals.

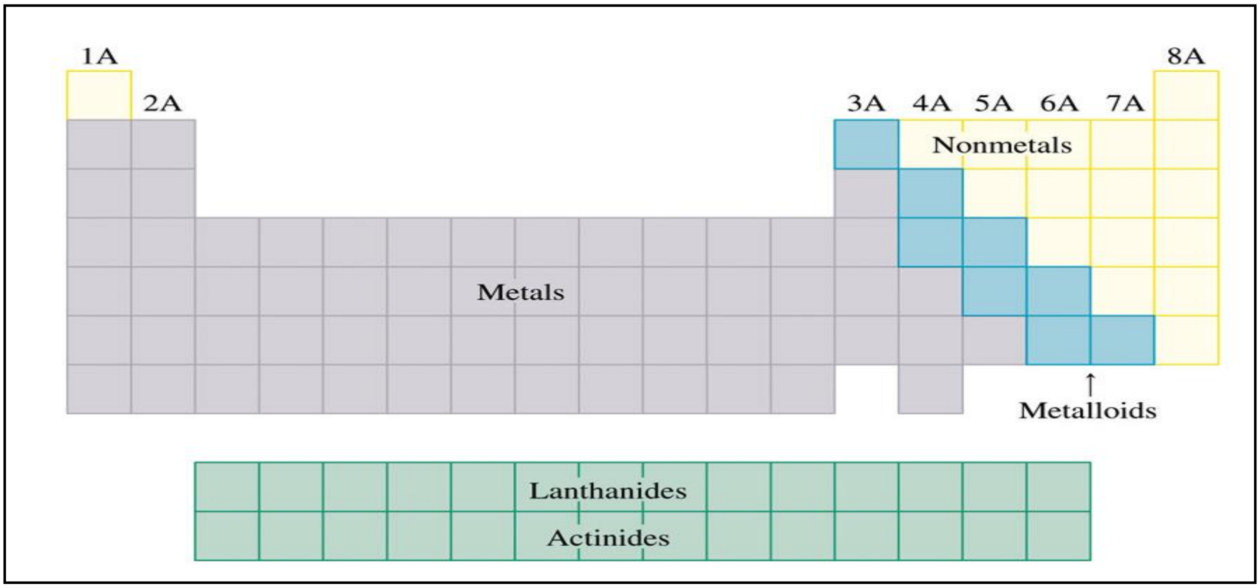
- The halogens, the members of Group 7A—fluorine (F), chlorine(Cl), bromine (Br), iodine (I), and astatine (At)—all form diatomic molecules. Fluorine, chlorine, bromine, and iodine all react with metals to form salts containing ions with a -1 charge (F-, Cl-, Br-, and I-).
- Noble gases The members of Group 8A—helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe), and radon (Rn). They all exist under normal conditions as monatomic (single-atom) gases and have little chemical reactivity.

A period: PeriodsThe horizontal rows of elements in the periodic table.

- Horizontal row one is called the first period of elements consisting of only hydrogen (H) and helium (He).
- The second period has 8 elements beginning with lithium(Li_3)and ending with neon(Ne_{10}).
- The third period also has 8 elements beginning with sodium (Na) and ending with Argon (Ar) and this followed by period four having 18 elements, beginning with potassium (K) and ending with krypton (Kr_{18}).
- The third period has 18 elements beginning with $_{19}$K-$_{36}$Kr.
- the fifth period also have 18 elements it beginning with rubidium (Rb) and ends with xenon (Xe). 37Rb-58Xe.
- the sixth period also have 32 elements it beginning with $_{55}$Cs-$_{86}$Rn.
- the seventh period also have elements it beginning with 87Fr-……. Incomplete.



Special names for groups in the periodic table



Metal:

- *good conductor of both electricity and heat*
- *forms cations and ionic bonds with non-metals*
- *high electropositivity and tendency to lose electrons*
- *form basic oxides and are lustrous, ductile or malleable*
- *have high densities than nonmetals..*
- *They have significantly high melting points and boiling points than nonmetals*.

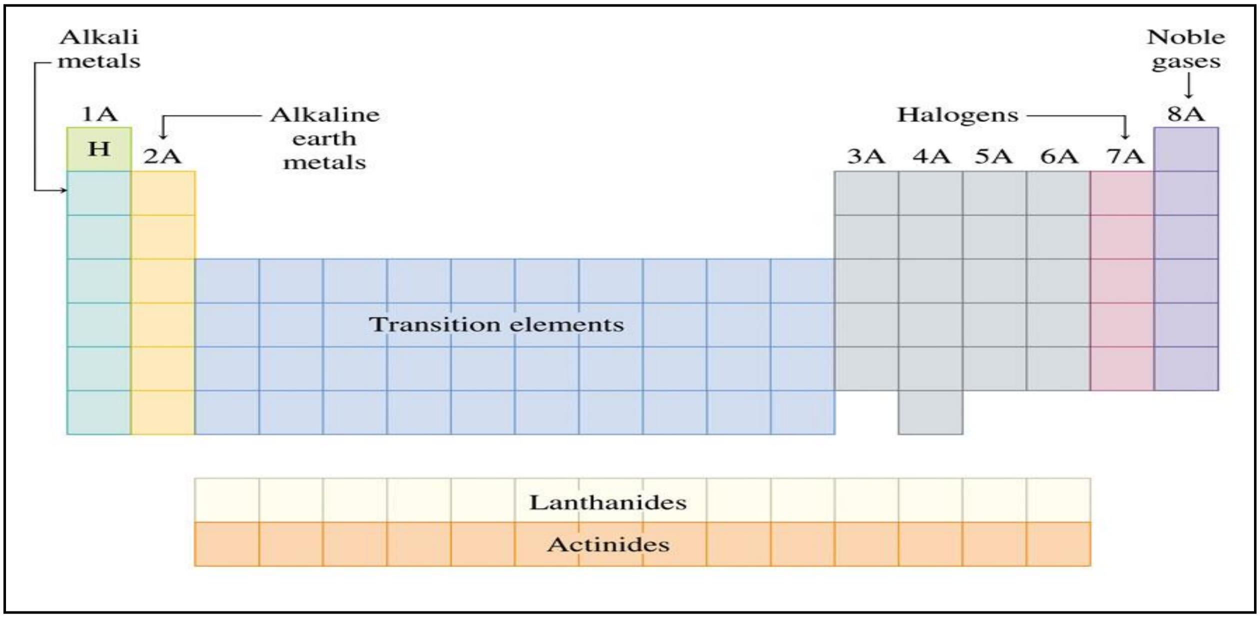
Nonmetal:

- *poor conductor of both electricity and heat*
- *forms anions and covalent bonds*
- highly electronegativity *and* tend to gain electrons forming negative ions
- form acidic oxides and in solid form , they are dull and brittle, rather than metals
- are usually having lower densities than metals
- They have significantly lower melting points and boiling points than metals (with the exception of Carbon).

Metalloids:

Are a few elements with intermediate properties (having both metallic and nonmetallic properties). The line that separates metalloids from nonmetals in the periodic table is referred to as the "amphoteric line". Metalloids often form amphoteric oxides. These elements, such as silicon (Si) and germanium (Ge) are usual ly good Semiconductors elements that, when pure, are poor conductors of electricity at room temperature but become moderately good conductors at higher temperatures.There are (B, Si , Ge , As , Sb, Te, Po).

Special names for groups in the periodic table



Transition metals:

referred to any element in the d-block of the periodic table, which includes groups 3 to 12 on the periodic table. All elements in the d-block are metals. In actual practice, the f-block is also included in the form of the lanthanide and actinide series. IUPAC definition states that a transition metal is "an element whose atom has an incomplete d sub-shell.

Inner transition metals: those elements whose atoms or ions have valence electrons in f-orbitals. There are two series of elements known as the lanthanoids (in which the electrons are in 4f orbitals)

and actinoids (in which the electrons are in 5f orbitals).

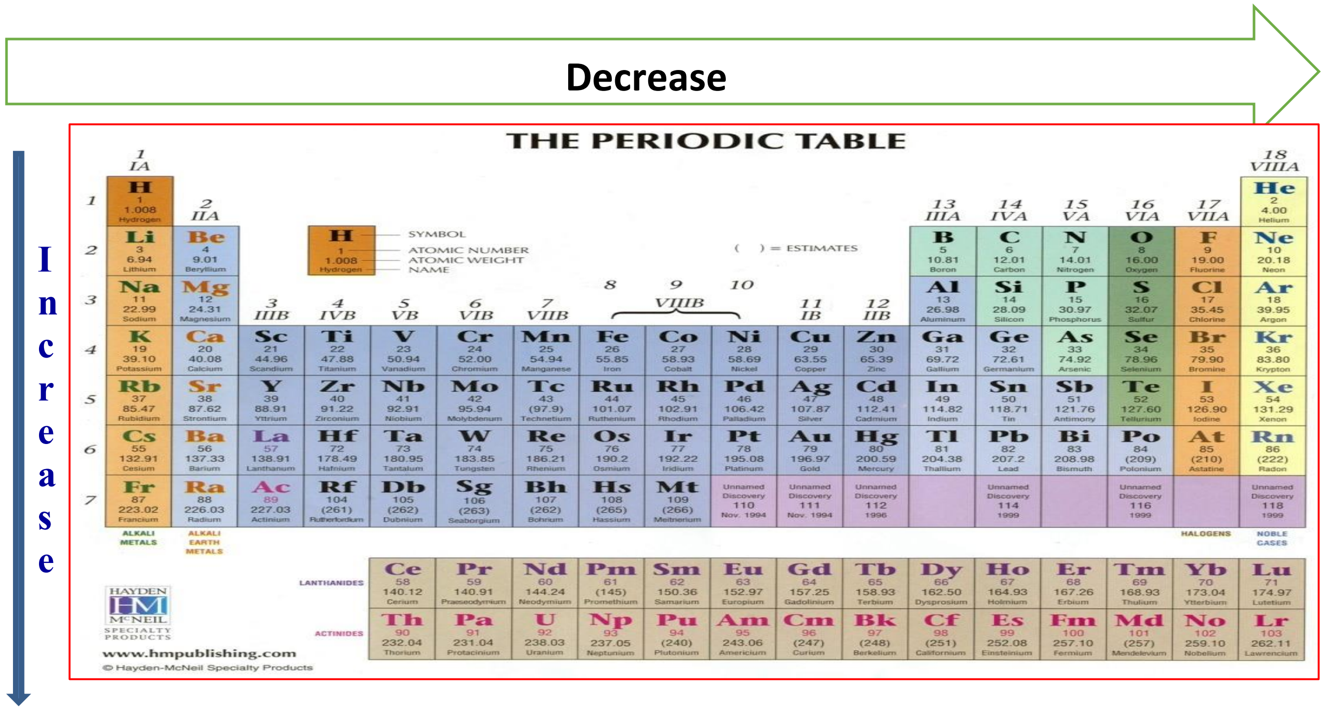
“Some Periodic Properties”

1-Atomic size (AS) and atomic radius (AR): It the distance from the center of a nucleus and the outermost shell of an atom. it is measured in angstrom A(1A = 1×10-10).

Atomic Radius; It is the half of distance between tow centers of two symmetrical neighboring atoms in a rigid and pure crystal(ionic radius) or in gaseous state(covalent radius).

Atomic size decreases from left to right in period and increases from top to bottom in a group.

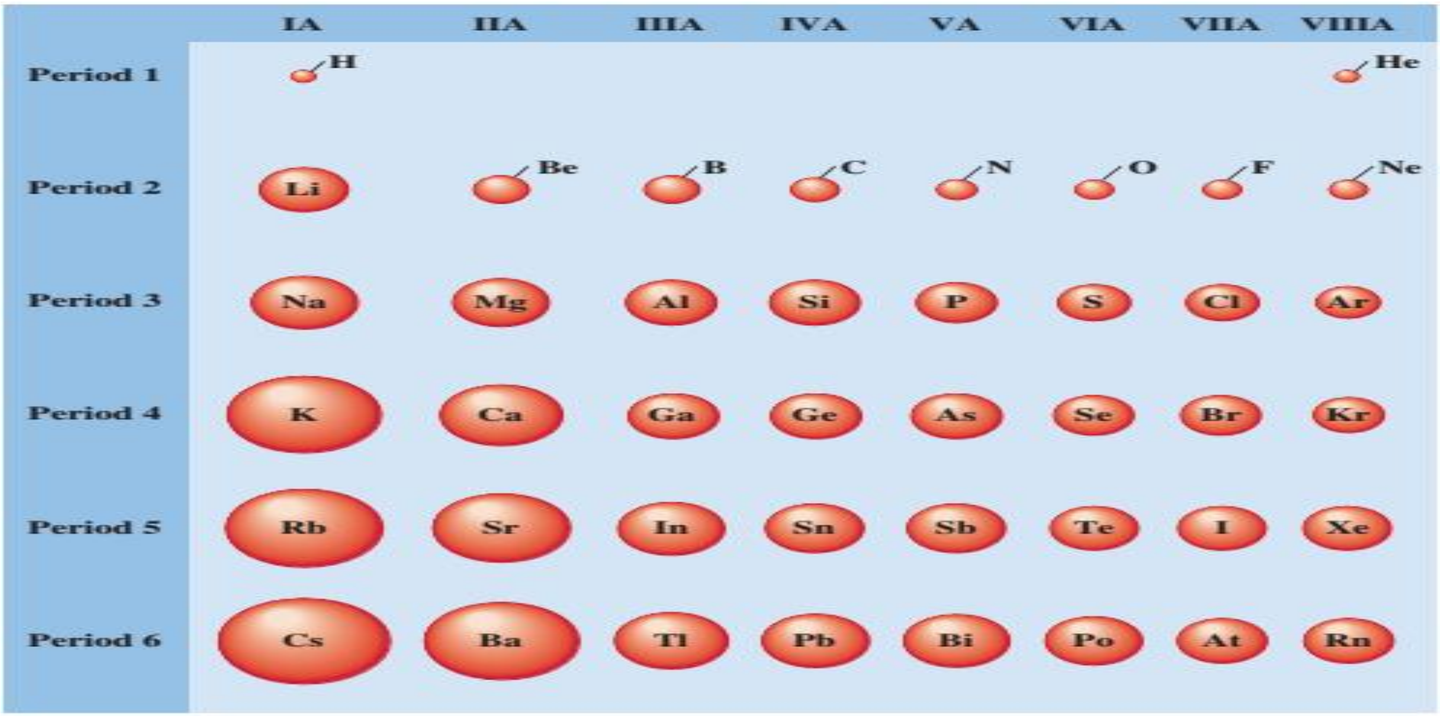
In a group like from hydrogen to lithium to sodium on down--the atomic size increases because there are more protons but there are more orbits of electrons in the atom so it takes up more



space with the extra electrons so as you go down a group, the size increases. As you go across a period, as from lithium to neon, the size decreases because there are more protons in the atoms but the electrons are all in the same orbit so they are pulled closer to the nucleus so the size smaller.

Within each period (horizontal row), the atomic radius tends to decrease with increasing atomic number (nuclear charge). The largest atom in a period is a Group IA atom and the smallest is a noble-gas atom.

Within each group (vertical column), the atomic radius tends to increase with the period table.



Size of Ions "Ionic Radius"

When atoms gain or lose electrons, the atom becomes an ion.

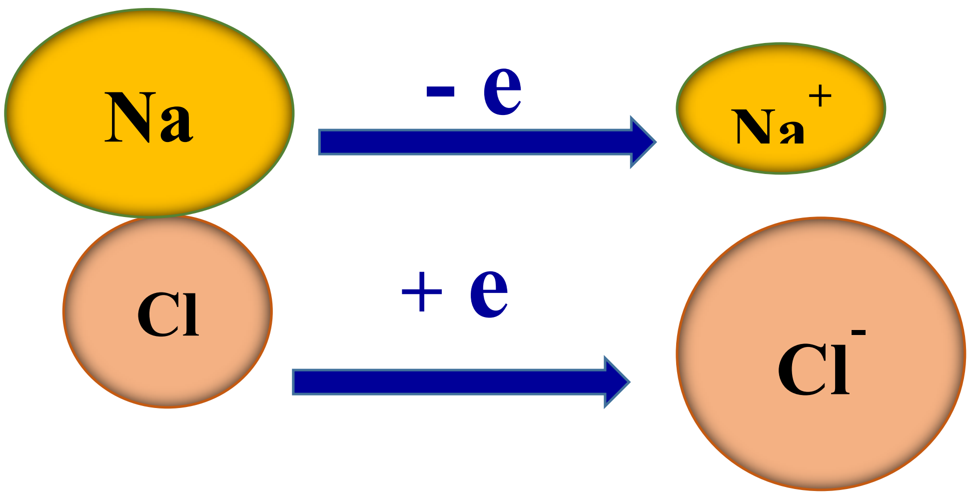
When an atom gains an electron, it becomes a negatively charged ion that we call an anion.

Anions are larger in size than their parent atoms because they have one or more additional electrons, but without an additional proton in the nucleus to help moderate the size.

When an atom loses an electron, it becomes a positively charged ion called a cation.

Cations are smaller than their parent atoms because they have lost electrons (sometimes the entire outermost energy level) and the electrons that remain behind are pulled closer to the nucleus

Size of Ions "Ionic Radius":

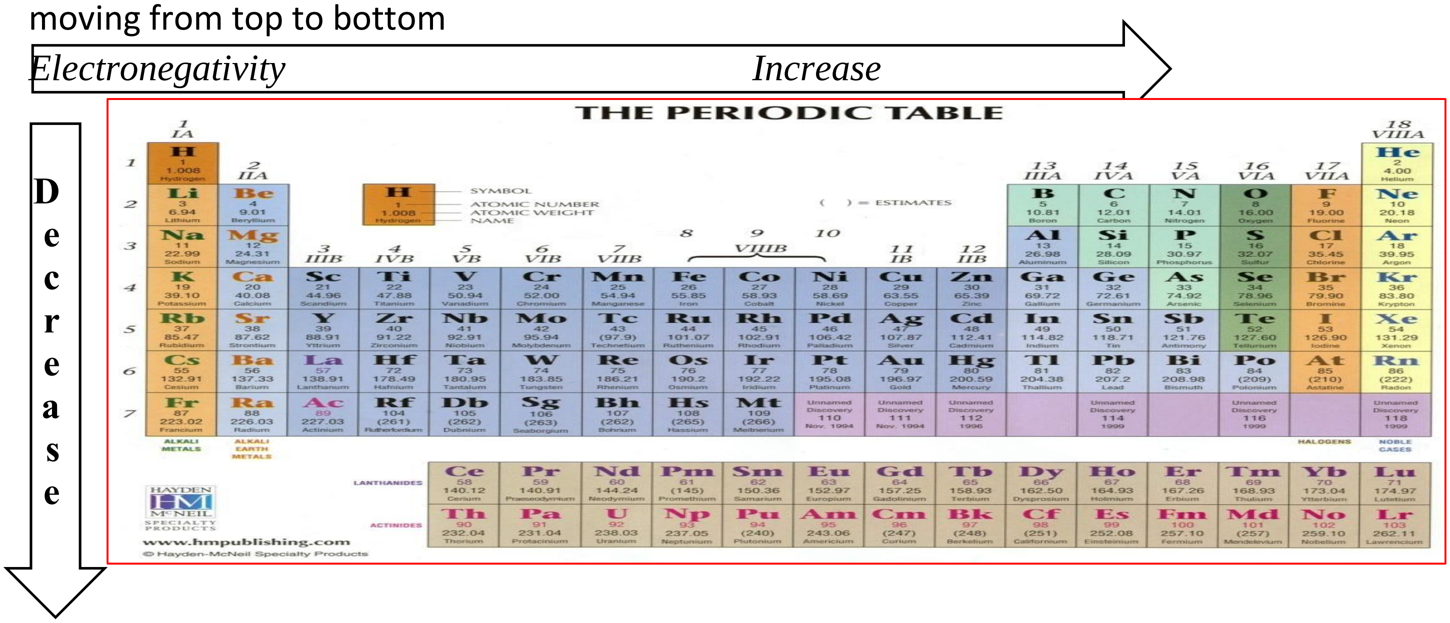


2- Electronegativity *(EN)* :

It is the tendency of an atom in a molecule to attract electrons to itself in a chemical bond.

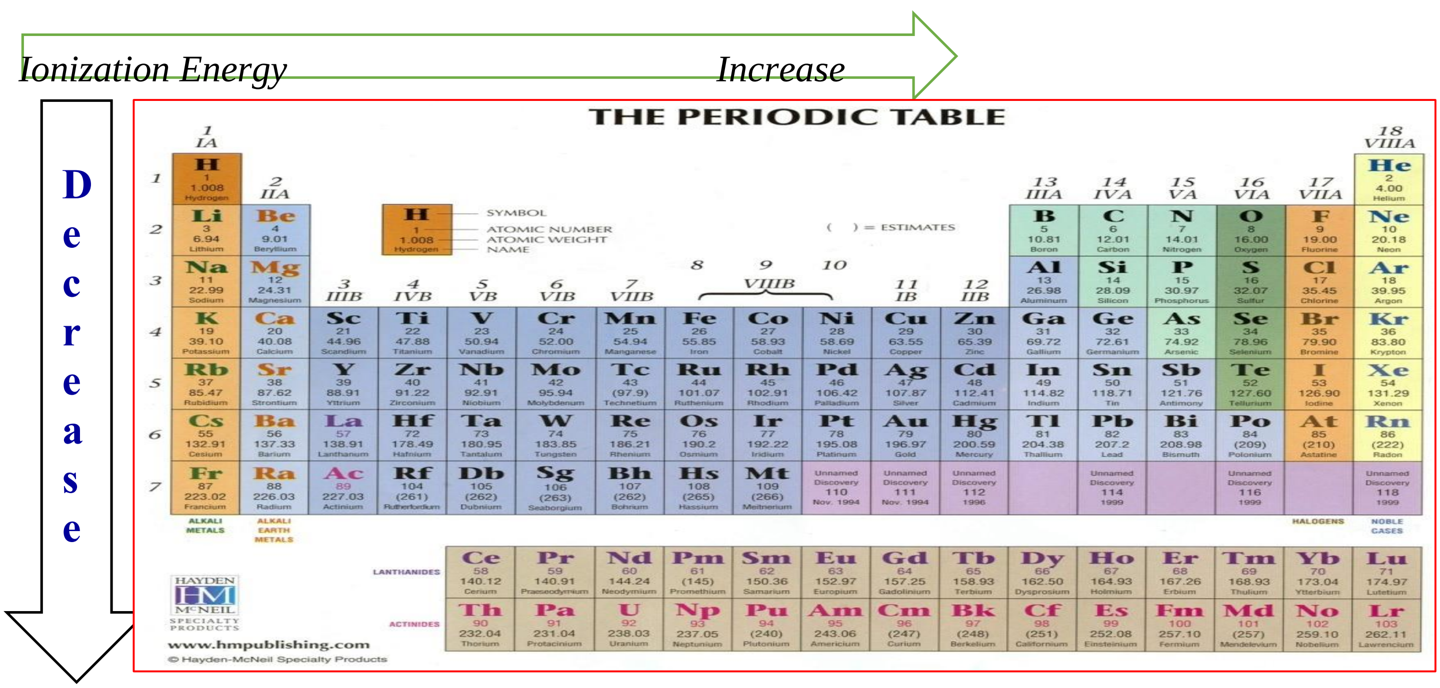
Small atoms are more electronegative than large atoms.

Electronegativity decreases in a period on moving from left to right and increases in a group on moving from top to bottom



3- Ionization Energy (IP) or ionization energy (IE):

It is the amount of energy required to remove the most loosely bound electron from an isolated gaseous atom. The energy required to remove the first electron is called the first ionization potential and that required to remove the second electron is called ionization potential.



First ionization potential. $M \rightarrow M^{+}$

Second ionization potential. $M^{+} \rightarrow M^{2+}$

Third ionization potential. $M^{2+} \rightarrow M^{3+}$

1^{st} I.E < 2^{nd} I.E < 3^{rd} I.E

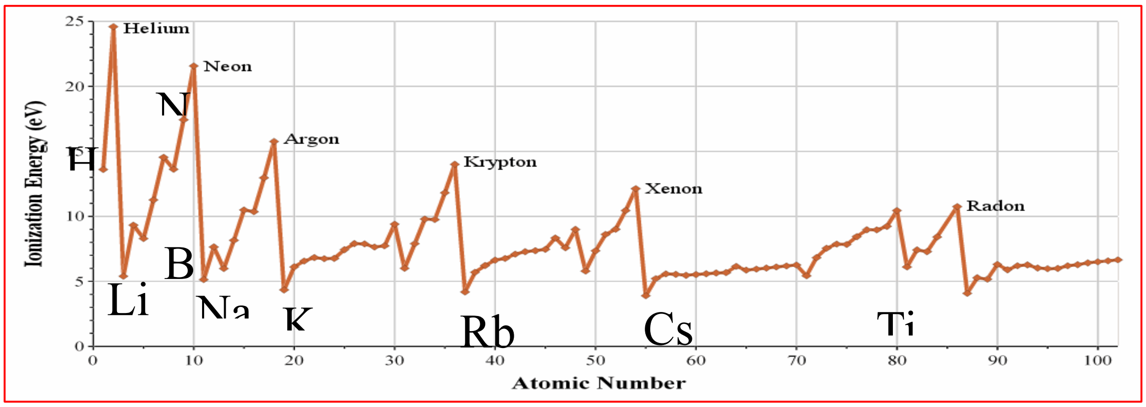
The Ionization Energy depends on:

1- Size of the atoms.

The I.E decreases as the size of the atom increases (Inversely proportional).

2- Charge on the nucleus. The I.E increases as the nuclear charge increases (Directly proportional). 3- Type of electron involved (Shape of the orbital).

3- Type of electron involved (Shape of the orbital).



The I.E depends on the type of electron which is removed. An s-electron is attracted to the nucleus more than p-electron in the same period.

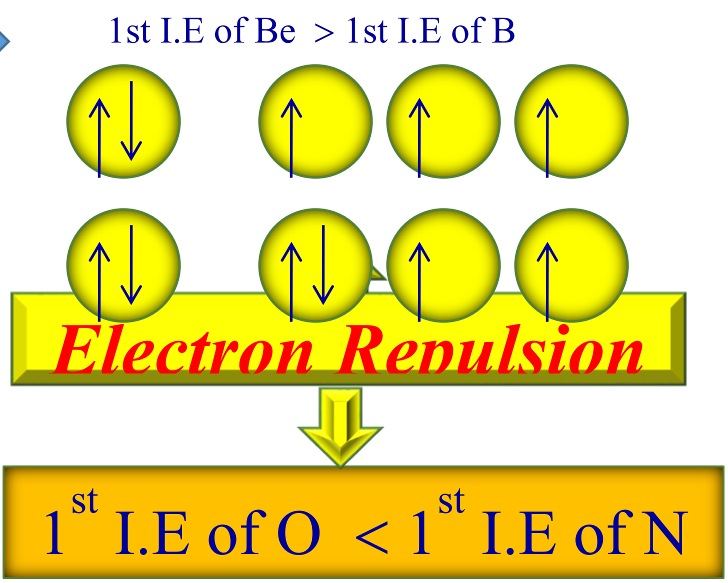
S > p > d > f

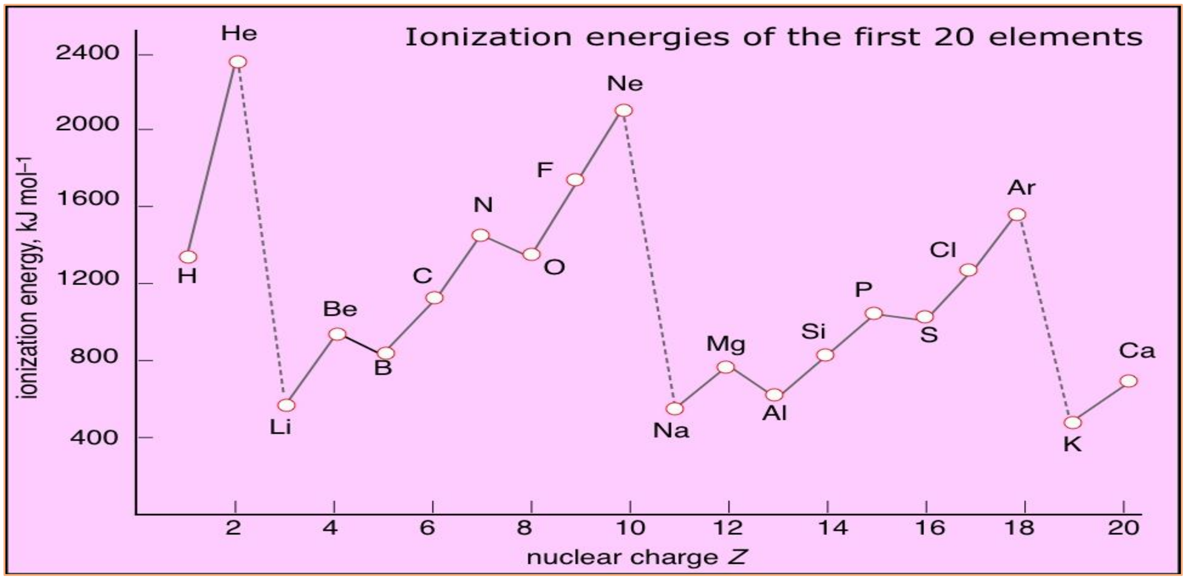
^{4}Be: $1s^{2}, 2s^{2}$

^{5}B: $1s^{2}, 2s^{2}, 2p^{1}$

^{7}N: $1s^{2}, 2s^{2}, 2p^{3}$

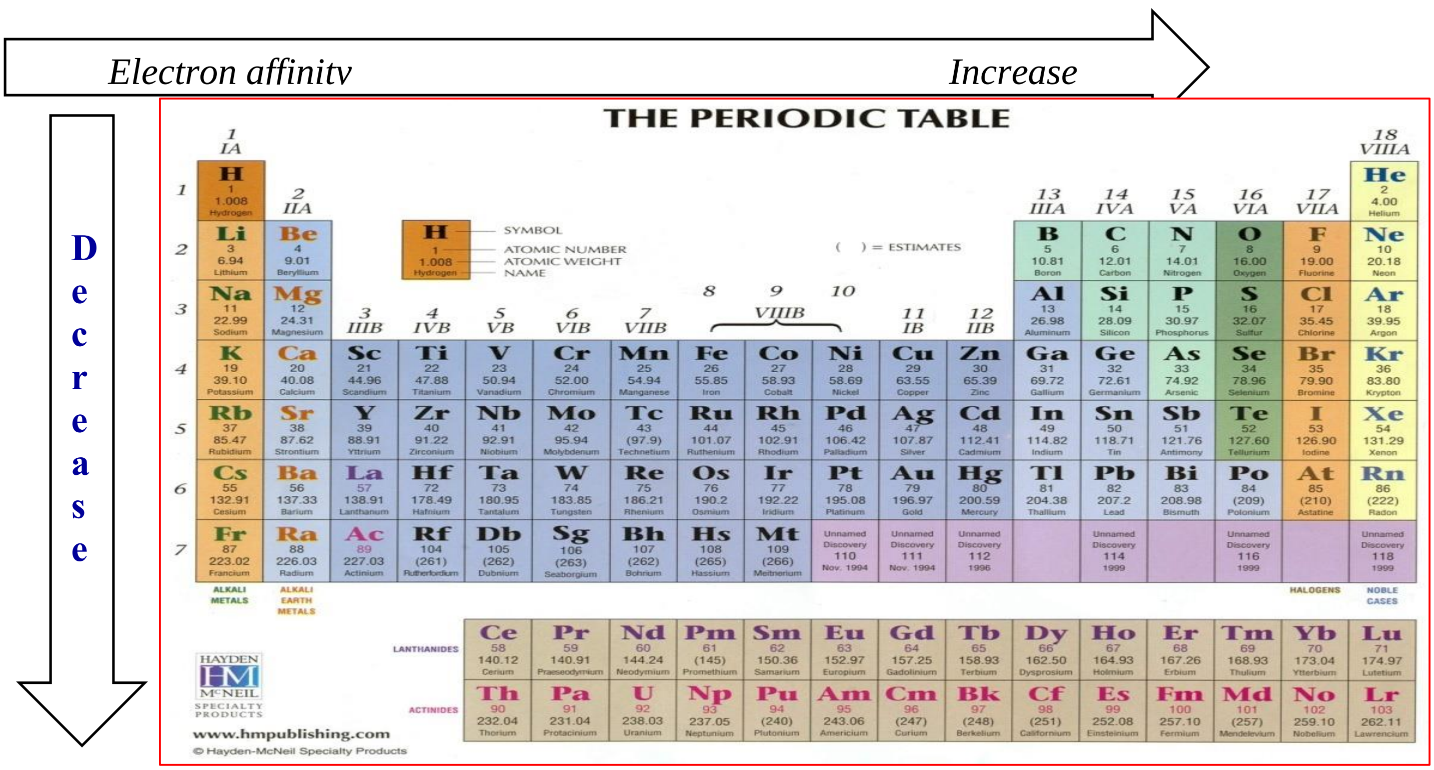
^{8}O: $1s^{2}, 2s^{2}, 2p^{4}$





4- Electron Affinity(*EA)*: it the amount of energy released when an electron is added to outermost shell of an isolated gaseous neutral atom. The electron affinity increases in a period on moving from left to right and decreases in a group on moving from top to bottom.

Ionization energy is concerned with positive ion, while electron affinity is concerned with negative ion. It is the energy change for the process of adding an electron to a neutral atom in the gaseous state to form a negative ion.



$X_{(g)}$ + e^- $\longrightarrow$ $X^-_{(g)}$ E.A. = + ΔH *Endothermic*

The positive values of electron affinities means that energy is required for the atom to become negatively charged.

$X_{(g)}$ + e^- $\longrightarrow$ $X^-_{(g)}$ E.A. = - ΔH *Exothermic*

The negative values of electron affinities means that energy is released when a gaseous atom accepts an electron.

E_{ea} generally increases across a period (row) in the periodic table. This is caused by the filling of the valence shell of the atom

A trend of decreasing E_{ea} going down the groups in the periodic table would be expected. The additional electron will be entering an orbital farther away from the nucleus. Since this electron is farther from the nucleus it is less attracted to the nucleus and would release less energy when added.

5- Metallic Character*(MC) or Electro positivity (EP):*

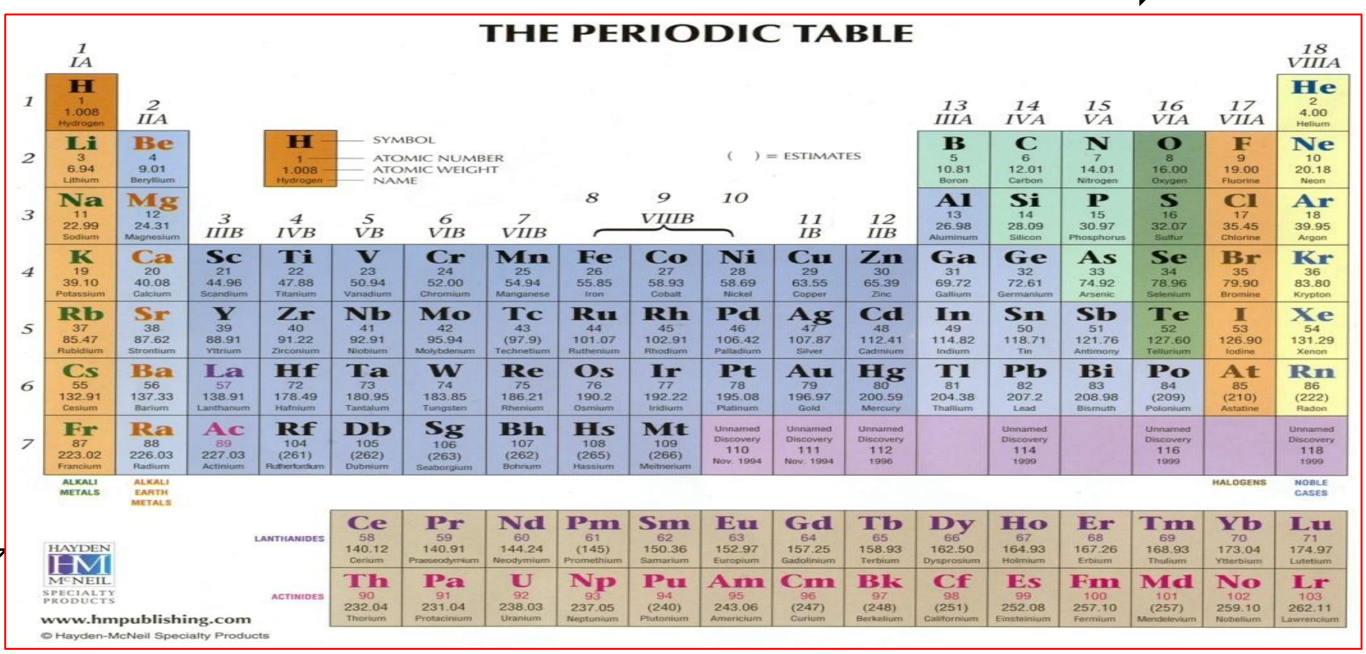
It is the ability of an atom to lose electrons if supplied with energy.

M $\longrightarrow$ M^+ + e^-

The electro positivity decreases in a period on moving from left to right and increases in a group on moving from top to bottom

Metallic Character *decrease*

increase



6-Non-metallic character (NC):: it is the tendency of an atom to gain one or more electron in its outermost shell to attain an octet of electrons. NC increases in a period on moving from left to right. It decrease in a group on moving from bottom to top.

The following chart summarizes the trends of properties across the periodic table: